Real Estate Price Prediction

https://github.com/aiyshwary/real-estate-price-prediction

Python | pandas | NumPy | scikit-learn | Matplotlib | Seaborn

OVERVIEW

A supervised regression case study that predicts house prices using the kc_house_data dataset, benchmarking multiple models and comparing performance with R2 and MSE metrics.

IMPACT

Benchmarked six regression models with bagging/boosting and hyperparameter tuning to identify the most accurate predictor for house prices.

TECHNICAL WALKTHROUGH

Project Contents

The project includes a Jupyter notebook, the kc_house_data.csv dataset, and a slide deck summarizing findings.

- Notebook: ML_HOUSE_RENT_PREDICTION_CASE_STUDY IPYNB (1).ipynb
- Dataset: kc_house_data.csv (house attributes and prices)
- Slides: ML GROUP 8 (1).pptx

Data Preparation

Loaded the dataset, checked distributions and missing values, and prepared features for regression modeling.

- Exploratory analysis with pandas, matplotlib, and seaborn.
- Basic cleaning to ensure consistent numeric inputs.

Models Evaluated

Benchmarked multiple regression models to compare predictive performance.

- Linear Regression, KNN Regressor, Decision Tree Regressor
- Random Forest, AdaBoost, Gradient Boosting Regressors

Training & Tuning

Applied bagging and boosting techniques, then used hyperparameter tuning to improve generalization.

- Model selection guided by R2 and MSE metrics.

Results

Compared model scores and selected the best-performing regressor based on error metrics.

Reproducibility
Notebook can be run end-to-end with the dataset placed in the project root, producing charts and model metrics.

KEY DETAILS

- Loaded and explored kc_house_data.csv, performing basic cleaning and feature analysis for regression modeling.
- Trained Linear Regression, KNN Regressor, Decision Tree, Random Forest, AdaBoost, and Gradient Boosting models.
- Compared model performance using R^2 and MSE to select the best-performing regressor.
- Applied bagging/boosting strategies and hyperparameter tuning to improve predictive accuracy.
- Documented the workflow in a Jupyter Notebook and included presentation slides for project reporting.